



Bobathon Grid Optimiser

Demand Spike Forecasting · Anomaly Detection · Operator Briefs with Bob AI

TEAM VRAM

15 SEPTEMBER 2026

The Problem: Grid Operators Are Flying Blind

Unpredictable Renewable Output

Solar and wind variability creates gaps in supply that are hard to anticipate

Sudden Demand Spikes

Rapid load changes strain the grid before operators can respond

Fragmented Signals

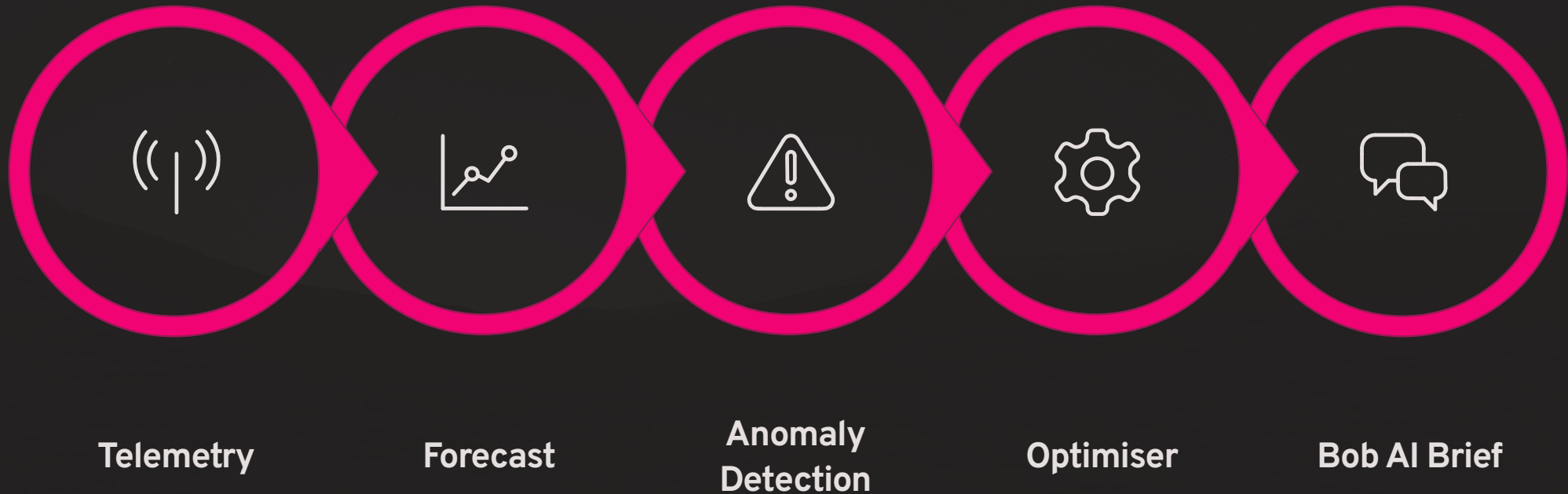
Telemetry, forecasts, and alerts live in separate systems – no unified view

⚠️ **Impact:** Increased curtailment, load-shedding risk, and costly manual triage.



The Solution: End-to-End Pipeline

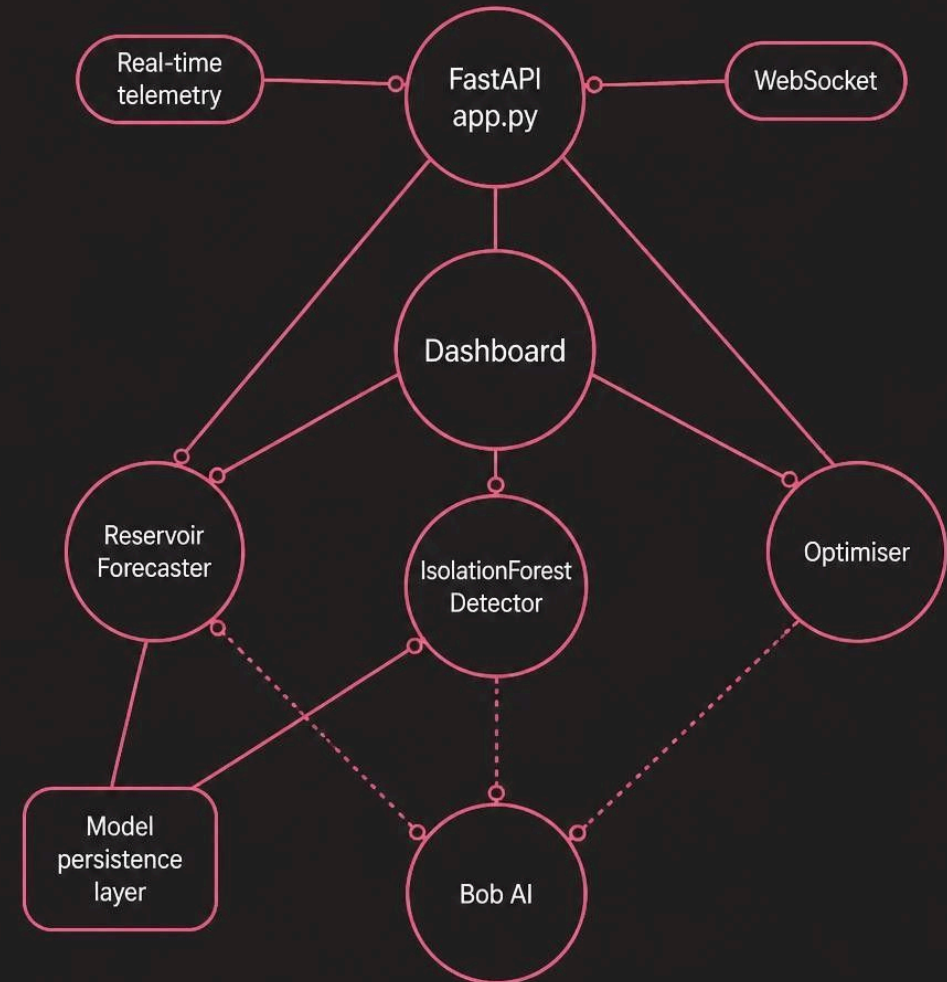
Raw grid telemetry flows through forecasting, anomaly detection, and optimization – then Bob AI generates a plain-language operator brief, all surfaced on a unified dashboard.



Each stage builds on the last – from raw signal to operator-ready decision support.

System Architecture

The pipeline is orchestrated through a **FastAPI backend** (app.py) that manages real-time telemetry via WebSocket, coordinates model inference, and persists trained models for low-latency serving.



Reservoir Forecaster

Demand spike prediction from streaming telemetry

IsolationForest Detector

Unsupervised anomaly flagging on grid signals

Bob AI Integration

bob_integration.py – briefs, fallbacks, diagnostics

Live Demo: Dashboard Walkthrough

1

Connect

Link real-time telemetry feed

2

Forecast

View predicted demand curve

3

Detect

Anomalies flagged automatically

4

Recommend

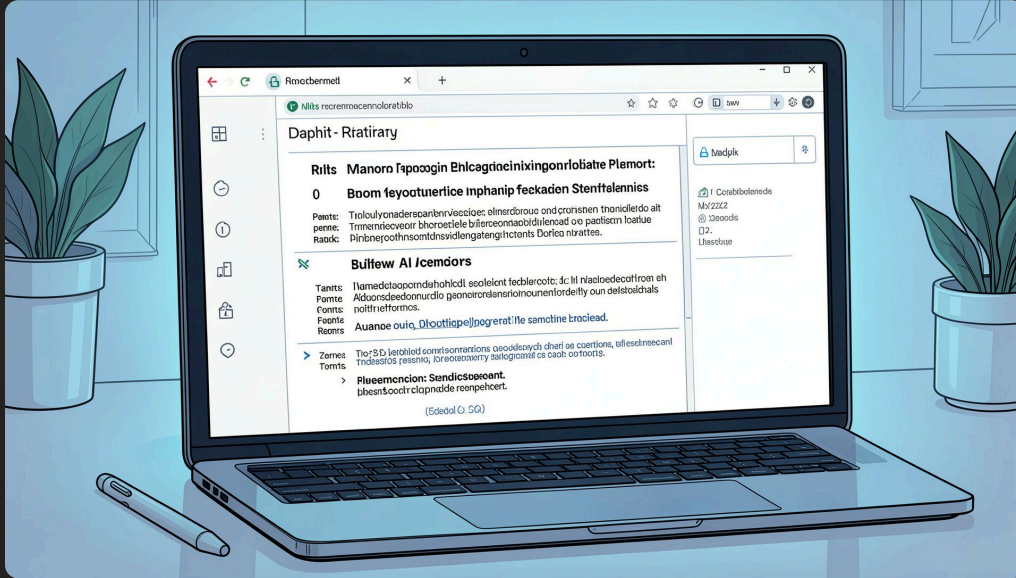
Optimizer suggests actions

5

Bob Brief

Plain-language operator summary

Bob AI: Operator Briefs That Explain Themselves



Bob AI translates model outputs into **actionable operator briefs** — explaining what happened, why it matters, and what to do next.

→ Contextual Explanations

Briefs reference specific anomalies and forecast deviations

→ Prioritized Actions

Recommendations ranked by urgency and impact

→ Resilient Integration

Fallback handling, diagnostics, and Cloudflare/API restriction management in `bob_integration.py`

Results & Impact



Forecast MAE

Evaluated on synthetic demand data to measure prediction accuracy



Note: Results are based on synthetic data. Real-world validation and safety constraints remain future work.



Inference Latency

End-to-end pipeline speed tested for real-time operator use



Anomaly Precision

IsolationForest evaluated for outlier detection quality



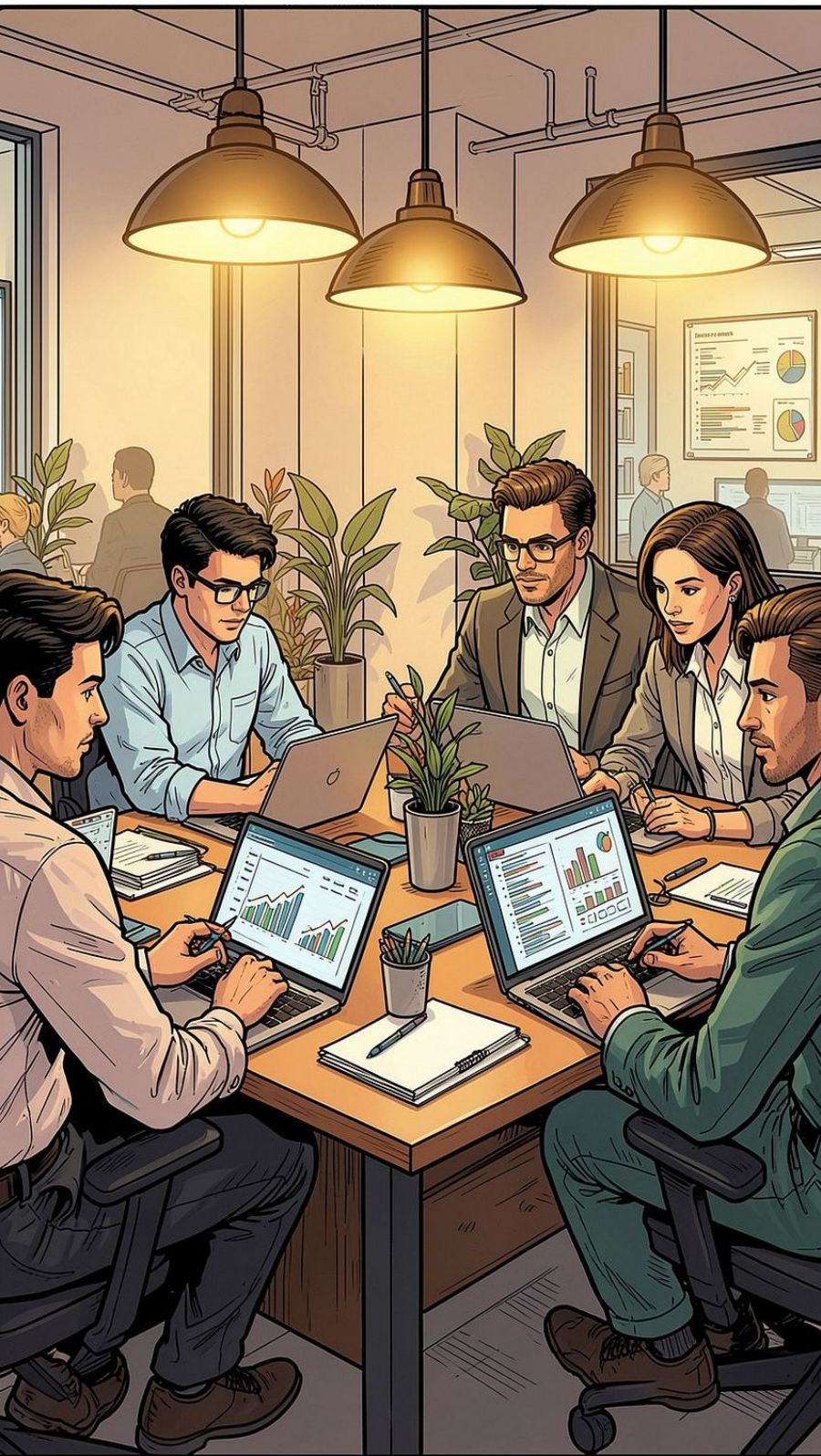
Faster Triage

Bob AI briefs reduce manual analysis time for operators



Reduced Curtailment

Forecasting enables proactive dispatch, lowering renewable curtailment potential



Team VRAM

Member 1 - Aastha Patel

Role & contribution

Member 2 - Meet K Patel

Role & contribution

Member 3 - Ruchit Jivani

Role & contribution

Member 4 - Vaibhavi Bhatt

Role & contribution

Forecast → Detect → Optimise → Explain

Thank You – Q&A